GRAINGUARD · PROJECT PLAN

GrainGuard project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 upgrade, stage P3)

GrainGuard project plan

GrainGuard is at stage P3 (design outputs) and TRL 3, the portfolio's caps. The next gate, P4 verification at TRL 4, is on hold until Amish lifts the cap. Three decisions are requested now: the concept choice (GGD-DEC-001 open item 10: all three options within 10 %, option A the design of record), the battery's disconnect setting against cold (open item 11, recommended: the battery maker's 65 % figure) and, when the cap lifts, the first contact with a land-grant extension grain-storage specialist to find the host farm. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Pipeline documents	Project lead	STM, OPT, URS, PRS, RTM, REG, FEA, RSK, MKT, PLN	Done at P3 (this session)
Design and calculations	Design lead	GGD-PRC-001, GGD-CAL-001, the D3 model of every option, STEP and STL, drawings GGD-DWG-001 and 101 to 111	Done at P3; open items 10 and 11 with Amish
Build plan	Design lead	GGD-BLD-001 with pictures from the model	Done at P3
Renders and storefront images	Project lead	Options picture, product renders, cards (on Amish's Mac)	Waiting for the next render session
Co-design partner	Project lead	A host farm with a small corn bin, a rated center hanger and an electrician, through an extension specialist	On hold (TRL 4 work)
Verification	Design lead	Bench and field tests of P4	On hold (cap)

Resources needed

Table 2. Resources (P4 and later are planned, not started; costs are estimates).

Resource	Deliverable	Effort	Estimated cost (USD)
Parts for one prototype and its receiver	A built kit to the build plan	One farm workshop, about two weeks	432.50 (BOM, indicative)
Local fabricator	The mast weldment	Half a day	To quote locally
Licensed electrician	The relay kit at the host farm's starter	Half a day	To quote locally
Extension grain-storage specialist	Review of the control rule and targets; host farm	A few hours of meetings	None expected (extension service)
Laboratory or meter moisture samples	The PR-002 comparison	One season of sampling	To assess
Hazardous-locations reviewer	Assessment of the pods and cable in a dusty bin (GGD-REG-001 A2)	A review	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Met
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT, REG complete	Met
P2 Design inputs	PRS with targets, tolerances and verification, CAL, RTM, concept selection DDR (GGD-DDR-004, awaiting Amish), STEM check (open: S17 with Amish, seven rows to measure)	Met on paper; concept choice open
P3 Design outputs	D3 model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Met (this session)
P4 Verification	Bench tests (relay fail-off, power budget, probe check), coated test board through a fumigation, moisture comparison, radio survey	On hold (cap)
P5 Validation	A season on the host farm with the farmer and the extension specialist	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Amish decides GGD-DEC-001 open item 10 (concept choice; no pick, all within 10 %) and open item 11 (disconnect setting; recommended the 65 % figure).
2. On Amish's Mac: the options picture (`python .kit/options.py`) and the product renders from `cad/src/product_model.py` .
3. When the cap lifts: contact a land-grant extension grain-storage specialist (GGD-DEC-001, item 4 of 2026-10-02), then the TRL 4 tests in the order of the build plan's first checks.

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